# Skills Summary

## Terms, Factors, Coefficients, and Degree

*Use this reference while completing your worksheet or when studying.*

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## Skill 1 — Read an area model as terms (area-model-to-terms)

**Model → notation:**
Side lengths are **factors**; each cell inside is one **term** of the product. Cells with the same power of $x$ are like terms and combine, so a four-cell model usually gives three terms.

$(x+p)(x+q) = x^2 + (p+q)x + pq$

**Example:** Expand $(x+2)(x+4)$ with a model.
**Step 1.** Each side is a factor, so I multiply every part of one side by every part of the other and read off the cells.
**Step 2.** Cells: $x^2$, $4x$, $2x$, $8$. The middle two are like terms ($4x + 2x = 6x$): $\boldsymbol{x^2 + 6x + 8}$

**Try it:** Expand $(x+1)(x+7)$.

check: $\boldsymbol{x^2 + 8x + 7}$

## Skill 2 — Name the factors of a polynomial (factors-from-product)

**Rule:** Factoring rewrites a *sum of terms* as a *product of factors* — the same value, a different form.

Always look for a common factor first: every term of $ax^m + bx^n$ shares the greatest common numerical factor and the lowest power of $x$. Degrees of factors **add** to the degree of the product.

**Example:** Factor $x^2 + 9x + 20$.
**Step 1.** The terms share no common factor, so I look for two numbers whose product is the constant term and whose sum is the middle coefficient.
**Step 2.** $4 \cdot 5 = 20$ and $4 + 5 = 9$, so $\boldsymbol{(x+4)(x+5)}$ — two factors of degree 1, product of degree 2.

**Try it:** Factor $4x^2 - 6x$ completely (pull out the greatest common factor).

check: $\boldsymbol{2x(2x-3)}$

## Skill 3 — Coefficient and degree (coefficient-and-degree)

**Vocabulary on one term, $-7x^3$:** coefficient $-7$ (the sign belongs to it), variable $x$, degree 3 (the exponent).

**Polynomial:** its **degree** is the largest term degree; its **leading coefficient** is the coefficient of that highest-degree term — found by writing the polynomial in standard form (degrees decreasing), not by reading the first term as typed. A constant has degree 0; $x$ alone has coefficient 1.

**Example:** Put $5 - 2x^2 + 7x^3$ in standard form; give its degree and leading coefficient.
**Step 1.** Leading coefficient is defined from standard form, so I reorder the terms by degree before reading anything off.
**Step 2.** Degrees are 0, 2, 3, so descending order gives $\boldsymbol{7x^3 - 2x^2 + 5}$ — degree 3, leading coefficient 7.

**Try it:** Write $3x - 8 + x^4$ in standard form. What is its leading coefficient?

check: $\boldsymbol{x^4 + 3x - 8}$, leading coefficient 1

## Skill 4 — Same expression, two forms (equivalent-forms)

**Rule:** Two expressions are **equivalent** when they give the same value for *every* $x$ — proved by algebra (distributing or factoring), never by checking a few inputs in a table.

Counting terms does not decide it: $3(2x+5)$ is one term made of two factors, and $6x+15$ is two terms; they are the same expression written two ways.

**Example:** Show $2(3x - 4)$ as a sum of terms.
**Step 1.** The expression is a product, so distributing the outside factor turns it into a sum without changing its value.
**Step 2.** $2 \cdot 3x = 6x$ and $2 \cdot (-4) = -8$, giving $\boldsymbol{6x - 8}$ — one term became two, the value did not change.

**Try it:** Expand $(x + 5)(x - 5)$. How many terms does the answer have?

check: $\boldsymbol{x^2 - 25}$, two terms